

## FACIAL SCANNER PROJECT REPORT

### 1. Methodology

The face unlock system operates through a modular and efficient pipeline that ensures fast, reliable, and lightweight face recognition suitable for real-time applications. The system is divided into three main stages: detection, embedding, and recognition.

First, a YOLOv8n model is used to detect faces in real-time from webcam frames. Once a face is detected, it is cropped from the original frame and passed to an embedder model based on ResNet, which generates a fixed-length feature vector (embedding) that uniquely represents that individual's facial features.

For registration, multiple such embeddings (typically 10) are averaged and saved along with the user's name in a local CSV file. During authentication, a fresh embedding is generated and compared against the saved embeddings using a K-Nearest Neighbors (KNN) classifier. If the most similar embedding crosses a confidence threshold, the user is authenticated. The entire flow is wrapped in a user-friendly Streamlit UI that offers two modes: registration and unlock.

### 2. Dataset Preparation

The system does not rely on an external dataset. Instead, it dynamically creates its dataset during user registration. When a new user opts to register, the webcam captures 10 samples of their face. Each face is automatically cropped using YOLOv8n and passed through the ResNet embedder to extract feature vectors.

These vectors are then averaged to produce a single, robust face embedding per user, which is stored in a CSV file (data/embeddings.csv) along with the username. This makes the system lightweight and privacy-focused, with no dependency on large or sensitive datasets. The face images themselves are not stored, ensuring data security.

### 3. Choice of Architecture

The architecture was carefully selected to strike a balance between speed, accuracy, and deployability on local devices:

**YOLOv8n:** We chose the nano version of the YOLOv8 model as the face detector due to its excellent real-time performance with minimal computational load. YOLO (You Only Look Once) is known for its single-stage detection pipeline, making it ideal for live webcam processing.

**ResNet-based Embedder:** For facial feature extraction, a custom or pretrained ResNet-18 or ResNet-50 is used. ResNet is a deep convolutional neural network that handles the vanishing gradient problem effectively through skip connections. It transforms the cropped face into a meaningful numerical embedding that represents the user's identity in a compact form.

K-Nearest Neighbors (KNN): This simple yet effective classifier was chosen for its ability to work well with a small number of known users. It compares new embeddings to saved ones and returns the closest match. For small to medium datasets, it offers both interpretability and speed.

This modular stack ensures that each component can be improved or swapped independently based on future needs (e.g., replacing KNN with cosine similarity or adding face alignment).

#### 4. Potential Real-World Applications

This facial recognition system has multiple real-world use cases due to its modularity, simplicity, and offline capability:

Device Face Unlock: It can serve as a secure, lightweight authentication method for personal computers and mobile devices, replacing password-based login systems.

Employee Attendance Systems: Offices can integrate it to monitor employee attendance with real-time face capture, reducing manual input and proxy entries.

Access Control in Secure Areas: Laboratories, data centers, or restricted zones can implement this to grant or deny entry based on face authentication.

Secure UI Logins: Any local application, such as a Streamlit dashboard or web UI, can use this system to authenticate users before allowing access to sensitive tools or analytics.

#### 5. Experimental Results

Threshold: A KNN confidence threshold of 0.7 was used to determine match validity.

Observations: Performance was best under well-lit and frontal conditions. Accuracy dropped slightly in low-light or occluded scenarios. Does not accurately capture required person's images when even the smallest images of other people in background is present.